COMPLEMENTARY FEEDING

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Objectives

1. To define complementary feeding
2. To understand guideline on complementary feeding
3. To underscore significance of complementary feeding
4. To highlight requirements and types of complementary foods.
5. To highlight complementary feeding in various stages of a child's growth.

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Complementary feeding

- Complementary feeding for infants refers to the **timely** introduction of **safe** and **nutritional foods** in **addition to breast-feeding** i.e. clean and nutritionally rich additional foods introduced at about six months of infant age

- CF should be timely, adequate, appropriate, and given in sufficient quantity
- It is well recognized that the period of 6-24 months of age is one of the most critical time in the growth of an infant)

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Definitions

Complementary food – Any nutrient-containing food or liquid that is given **along with breast milk** during the period of complementary feeding.

Transitional foods – Complementary foods especially designed to meet the nutritional needs of young children.

Family foods – Complementary foods similar to those consumed by the rest of the family.

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Best practices in complementary Feeding (1)

1. Exclusive breastfeeding for six months.
2. Continued frequent on-demand breastfeeding until two years of age or beyond.
3. Practice responsive feeding applying principles of psychosocial care.
4. Practice good hygiene and proper food handling

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Complimentary feeding

Breast milk ceases to give adequate nutrients to babies after 6/12. Thus **nutritional needs** derived from complementary food.

This period of complementary feeding is when malnutrition starts in many infants, contributing significantly to the high prevalence of malnutrition in children under five years of age worldwide.

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Why NO introduction of complementary feeding before 6 months?

- **GIT function(digestion and absorption):** carbohydrates and fats absorption is suboptimal in the first 6 months but improve as the baby grows.
- **Renal maturation:** limited ability to excrete excess fluids- breast milk has a low solute load to the kidney.
- **Oral-motor development:** 0-6 months designed for liquid feeding and breast milk is adequate. Jaw and tongue movement and appearance of teeth in the 2nd half of life equip the baby for foods other than milk.

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Feeding practices

- Inappropriate feeding practices are often a greater determinant of inadequate intakes than the availability of foods in the households.

Age	Frequency of complementary feeds	Breastfeeding
6- 8months	2-3 times	
9-11 months	3-4 times	Breastfeeding
12-24months	3-4 times PLUS 1-2 nutritious snack	

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Complementary feeding should adhere to these:

a. **Timely** – Give when the need for energy and nutrients exceed that provided via exclusive breastfeeding.

b. **Should be Adequate** – Provide sufficient energy, macro- and micro-nutrients (balanced).

c. **Should be Safe** – Hygienically prepared, fed and stored.

d. **Should be Appropriate** – In consistency depending on the baby's age.

e. **Should be consistent** – Child fed once she/he has signals of appetite and satiety(responsive feeding).

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Early complementary feeding? NO!!!

Why?

1. Malnutrition (under or over nutrition can occur)
2. Micronutrient deficiency
3. Incomplete digestion/ constipation
4. Physiological stress
5. Baby may refuse to eat

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Spectrum of Complementary foods:

a) Carbohydrates – Cereals, bread, potatoes, bananas
b) Proteins – Meat(red/white) and meat products, pulses(peas, beans, lentils)
c) Vitamins – Fruits and vegetables
d) Calcium – Milk, cheese, yoghurt
e) Iron – Red meat, chicken, fruits, vegetables
f) Snacks – Yoghurt, sliced/chopped fruits. **(NB:NO SWEETS, CHOCOLATES, BISCUITS, CRISPS, etc).** Help top up energy in between feeds

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Common plant protein sources (per 100g)

	Protein(g)	Zinc(mg)	Iron(mg)	Energy(cal)
Beans	8	1	2	112
Peas (green)	4	1	1.3	59
Cowpeas	7	15	2.4	134

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Common carbohydrate sources (per 100g)

	Energy(kcal)
Potato	70
Rice	384
Yam	98
Maize	396
Wheat	375
Cassava	133
Millet	395

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Common Vegetables Sources and Vitamins content(per100g)

	Vitamin A(IU|)	Vitamin C(mg)
Carrots	17033	8.4
Pumpkin	2271	9.0
Spinach	17707	28.1
Tomato	21901	22.7
Sweet potato- check yellow sweet potatoes	1025	19.1
Green beans	1282	16.3

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Common Fruits and vitamin contents (per100g)

	Vit C(mg)
Apple	5.7
Watermelon	9.6
Lime	29.1
Tangeine	30.8
Lemon	53.0
Papaya	61.8
Orange	53.2
Banana	9.1
Pear	4.0

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Feeding History Job Aid, children 6–23 months

- Is the child still breastfed?

— How many times per day? Day and night?
— If using expressed breast milk, how is the milk stored and given?

- What other foods is the child receiving?

— How many meals and snacks each day?
— How much food at each meal?
— What is the consistency of the main meals?
— Do meals include: animal-source foods, dairy products, dark green vegetables or red or orange fruits or vegetables, pulses (beans, lentils, peas, nuts), oil?
— Who helps the child to eat?
— What bowl does the child get food from (his or her own bowl, or the family pot)?
— Is the child given any vitamin or mineral supplements?
— How does the child eat during sickness?

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Responsive feeding (active feeding)

- Follows the principles of psycho-social care.
- The adequacy of complementary feeding (adequacy in short for timely, adequate, safe and appropriate) not only depends on the availability of a variety of foods in the household, but also on the feeding practices of caregivers.
- Feeding young infants requires active care and stimulation
- The caregiver is responsive to the child clues for hunger and also encourages the child to eat.

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Responsive feeding

- feed infants directly and assist older children when they feed themselves, being sensitive to their hunger and satiety cues;
- feed slowly and patiently, and encourage children to eat, but do not force them;
- if children refuse many foods, experiment with different food combinations, tastes, textures and methods of encouragement;
- minimize distractions during meals if the child loses interest easily;
- remember that feeding times are periods of learning and love - talk to children during feeding, with eye to eye contact.

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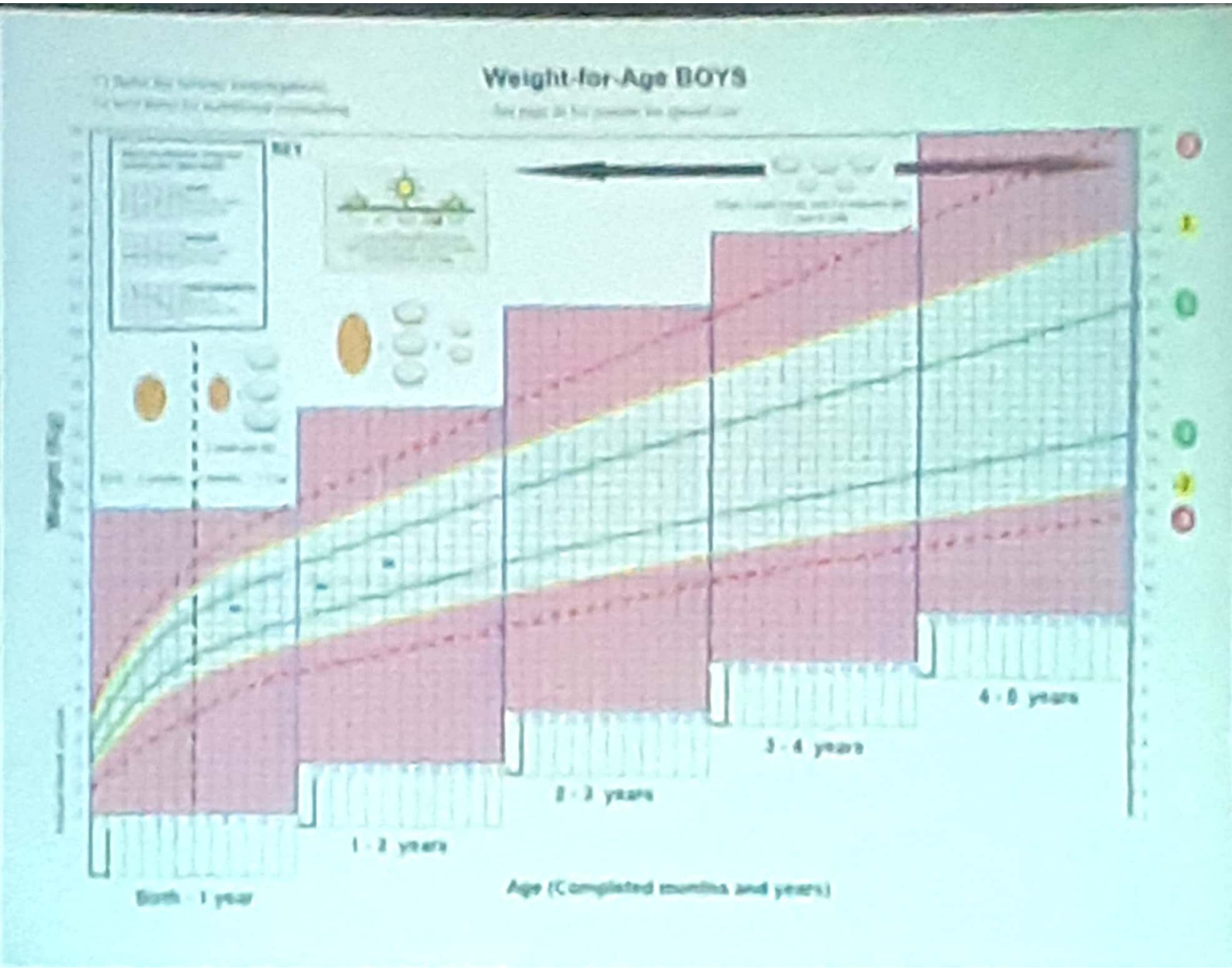

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Summary

- Do not Introduce complementary feeding early
- Increase the quantity as child grows
- Increase frequency of feeding as child grows
- Increase the food consistency and variety gradually as child grows.
- Diversify! Diversify! To improve on quality and the micronutrients.
- Embrace active feeding
- Good hygiene

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